I. ABSTRACT

Please provide a brief description of your proposed undergraduate research project, including a statement of the research problem, major hypotheses or questions posed in the research, the project's significance, and the methods used to examine the issue. Please include a statement of the theoretical and/or practical contributions and applications of your project.

We denote the initials as follows:

MC	Main Computer
CM	Microcontroller responsible for bot controls.
EM	Microcontroller responsible for the bot-side power system.
DP	Microcontroller responsible for the dock-side power system.

The transition towards smart campuses highlights the demand for autonomous service robot systems. For this study, we focus on an autonomous robotic system in a library setting, named ALAR, or A Library Assistant Robot, to handle repetitive tasks, such as book transportation and interacting with and assisting library users.

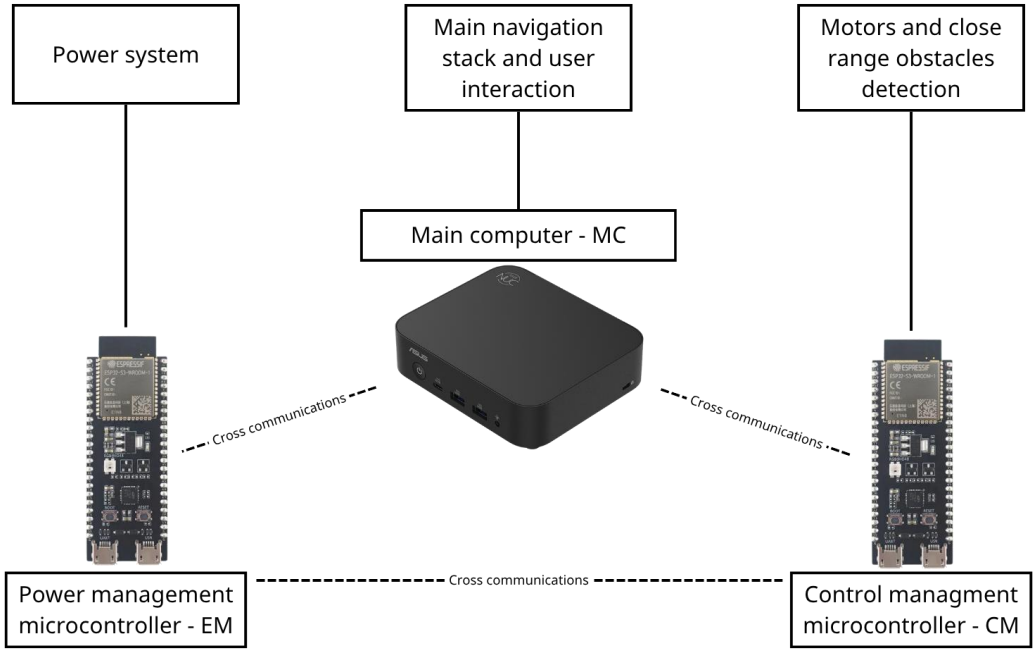
However, building such a robot system (including the robot itself and charging infrastructure) from the ground up poses a formidable engineering challenge for an undergraduate team. The primary research problem lies in the navigation algorithm, as well as in designing and integrating the electrical and hardware architecture. Safely operating and recharging a robot capable of navigating a dense, unpredictable environment requires a highly complex navigation, control, and power system. This complexity significantly elevates the risk of system failure, algorithm edge cases, power system short circuits, and so on. Compounding these technical hurdles are severe practical constraints, specifically the difficulty of sourcing specialized robotics components and the mechanical challenges of creating a compact, durable, and safe chassis with limited manufacturing resources.

This project poses the core question: *How can we engineer a safe, reliable hardware architecture for an autonomous robot system under budget, component limitations, and a dynamic environment such as a library?* We hypothesize that adopting a modular, decentralized system and an overkill power system design. The control system gives the system the ability to cross-check, provide redundancy, and communicate with each other in real time, and will be capable of shutting the most dangerous system down, creating a controlled failure environment. The power system design also follows the same philosophy of having human safety at the core; We aim to create a power system that can theoretically endure up to 500W of continuous DC power and be safe in the

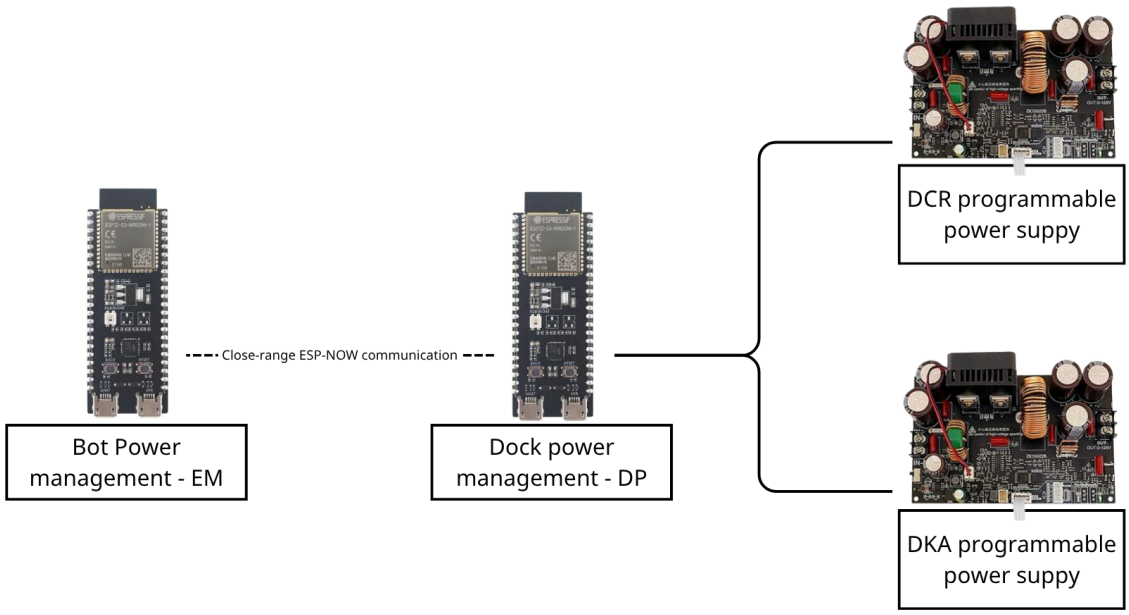
extreme event of short circuits.

Methodology:

Hardware: The project employs a distributed control strategy. Our main control computer, denoted as MC, will handle computationally intensive tasks - such as SLAM (Simultaneous Localization and Mapping) and obstacle avoidance using cameras - while 2 microcontrollers (denoted CM and EM for control management and energy management, respectively) will ensure real-time, low-level motor and relay control, power system monitoring, and serve as the 2nd and 3rd layer to verify total system status. Another microcontroller, denoted DP, will monitor the dock power system, control charging protocols, safeguard the charging system, and continuously communicate with EM on the robot. The hardware build will undergo iterative testing to ensure navigation and electrical safety, as well as structural integrity.



Bot side control system diagram



Dock side control system diagram

Software: The project utilizes ROS2, a Linux-based operating system for robotic applications, alongside dedicated ROS2 open-source add-ons and a mix of experimental navigation algorithms.

Ultimately, this project will deliver a fully functional prototype to further enhance library operations at VGU. Furthermore, its major theoretical and practical contribution is a validated, open-source-style engineering blueprint. This blueprint will demonstrate how to safely build high-power autonomous robots using accessible, off-the-shelf components, serving as a vital reference for future student researchers facing similar hardware bottlenecks.

Literature Review:

Recent advancements in campus automation emphasize the growing need in the development of various mobile robots for indoor services such as education and libraries [1]. The majority of the established solutions make use of ROS 2 middleware paired with LiDAR and visual SLAM to enable the dynamic localization and obstacle avoidance of the robot [2], [3]. In particular, the design of the distributed control architecture is gaining more and more traction in the robotics community as an alternative to a monolithic approach, which can introduce unacceptable delays in safety-critical systems [4]. Modern projects address both the high-level decision-making process implemented in ROS 2 on host computers and the low-level motor controllers and perception sensors [3]. The concept of autonomous docking is already well-established, with many current projects utilising visual markers or IR-based positioning [5]. However, the majority of the prior art only addresses the first part of the system, whereas a multi-stage power handshake and a redundant power supply to support high-current charging are two relatively unexplored areas for much-needed budget-friendly mobile robots. The proposed ALAR project aims to tackle the

problem by implementing a modular and fault-tolerant arrangement of the low-level microcontroller paired with a complex power circuit to ensure uninterrupted library operation.

References:

- [1] B. A. Croft and M. H. Smith, "Autonomous service robots in public and academic facilities: A review," *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 3, pp. 1120–1135, 2021.
- [2] S. Macenski et al., "Robot Operating System 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, p. eabm6074, 2022.
- [3] J. Zhang and S. Singh, "LOAM: Lidar Odometry and Mapping in Real-time," in *Robotics: Science and Systems (RSS)*, 2014.
- [4] M. Quigley, "Distributed embedded control architectures for mobile service robotics," *Journal of Intelligent & Robotic Systems*, vol. 84, no. 2, pp. 145–158, 2016.
- [5] M. C. Silverman et al., "Autonomous docking for dynamic battery recharging of mobile robots," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2002, pp. 1055–1060.
- [6] H. Park and K. Kim, "Fault-tolerant power management and dynamic charging protocols for indoor autonomous robots," *IEEE Access*, vol. 9, pp. 45210–45222, 2021.

II. DETAIL PROJECT PROPOSAL

Please describe your detail proposal below or submit it as a separate document.

Research questions:

This research focuses on the development of an autonomous library assistance robot system that can safely navigate the library environment, interact with and assist users, and maintain stable and electrically safe operation during continuous use. The scope of this research focuses on the development and evaluation of an autonomous library assistance robot for indoor library environments. The study covers autonomous navigation, obstacle avoidance, user interaction and assistance, as well as system stability and electrical safety. The robot's performance will be evaluated based on navigation speed, accuracy, obstacle avoidance, response time, operational stability, and user satisfaction.

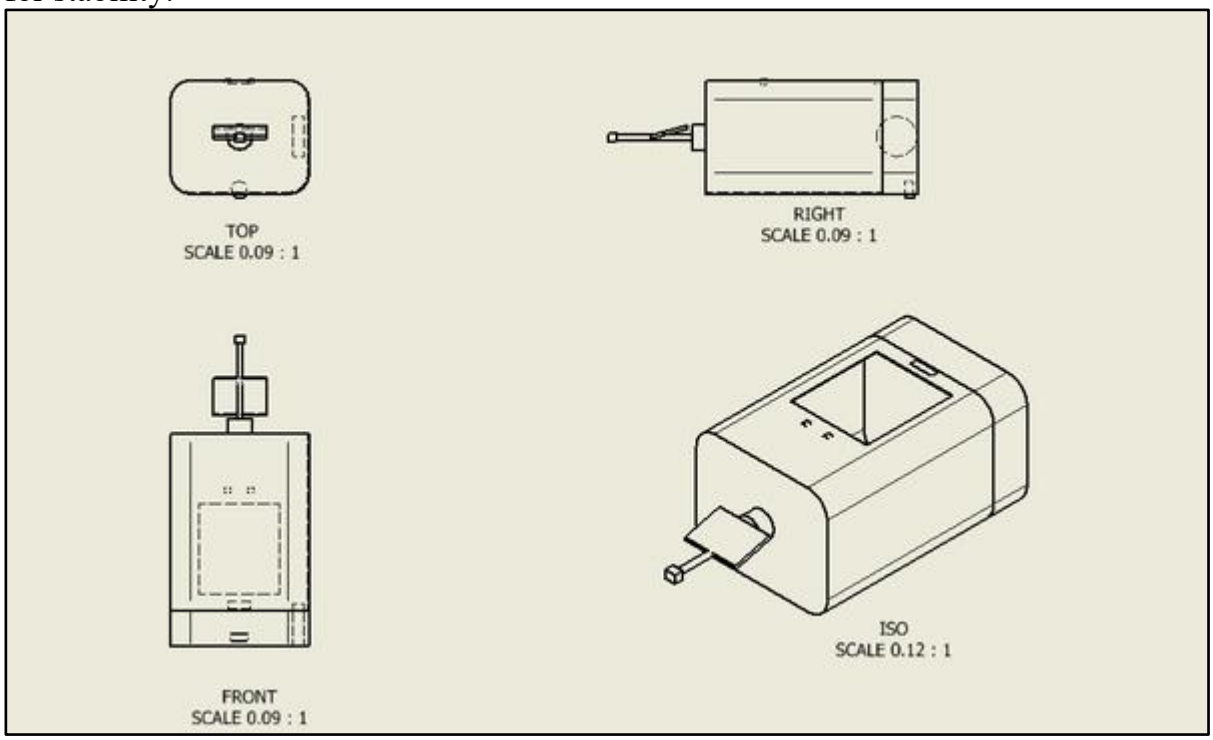
Project Objectives:

Overall, it is an autonomous library assistance robot system that can safely navigate the library and effectively assist and interact with library users with minimal human intervention.

Develop a safe autonomous navigation system that allows the robot to move efficiently in different areas of the library while considering the presence and movement of users and user-assistance functions, such as guiding users to specific books, shelves, or areas within the library. Human detection and human–robot interaction so that the robot can recognize users, respond to their requests, and provide basic assistance.

Design:

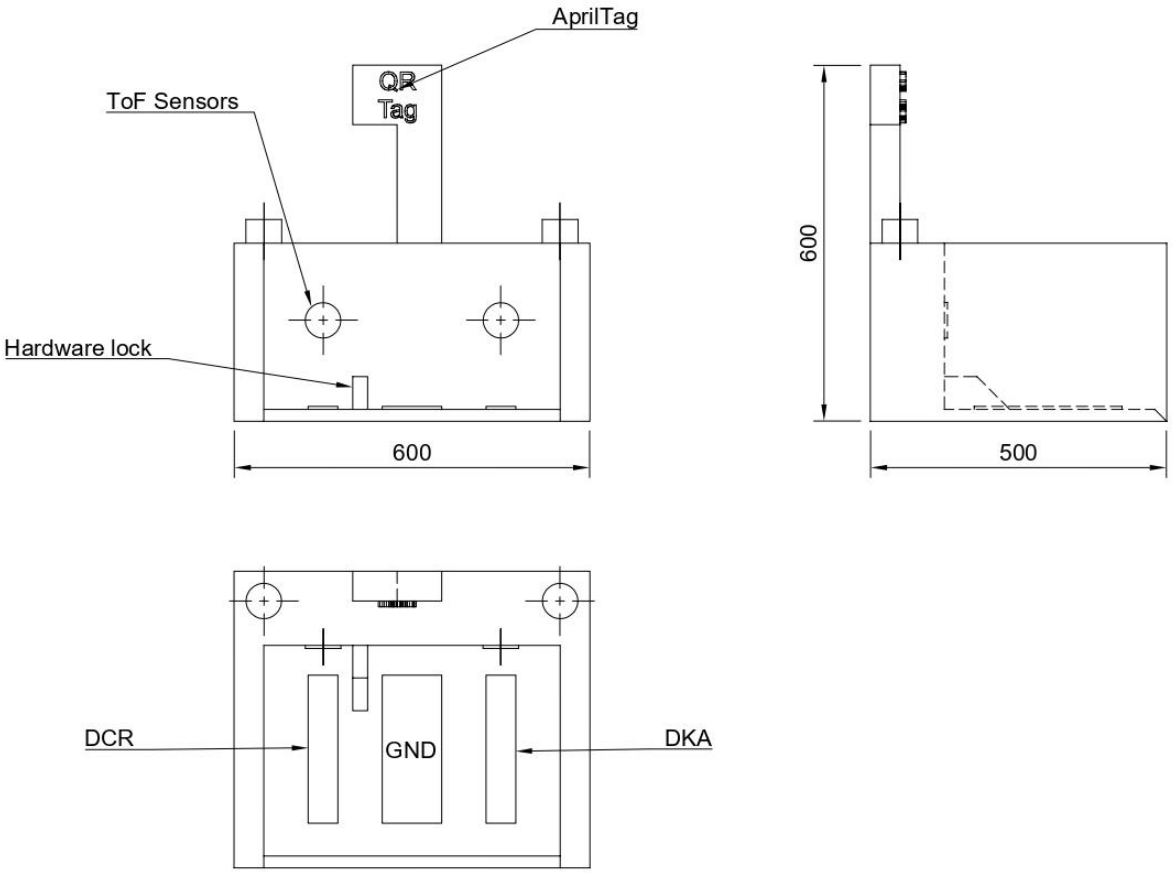
The mechanical design is divided into two main sections: a rounded rectangular base (400mm x 400mm, filleted corners for safe operation around people). This lower section houses the battery, motors, close-range obstacle sensors, and power distribution board. Above it will be a cylindrical upper section with 1.1m height. The cylindrical upper section has a top cap to mount the LiDAR sensor, a user-interactable display, as well as navigation and user-interaction cameras. A rectangular gap in the middle of the chassis serves as a multi-purpose container. Two stepper motors drive two main wheels and two extra omni wheels for stability.



ALAR design mockup

ALA's dock design

Power:

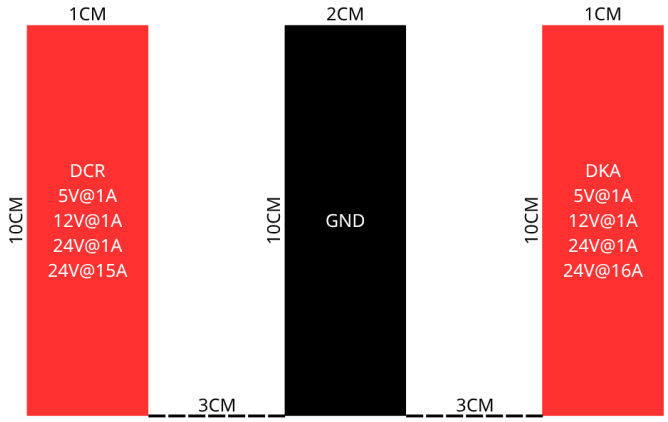


For the power section, we denote the initials as follows:

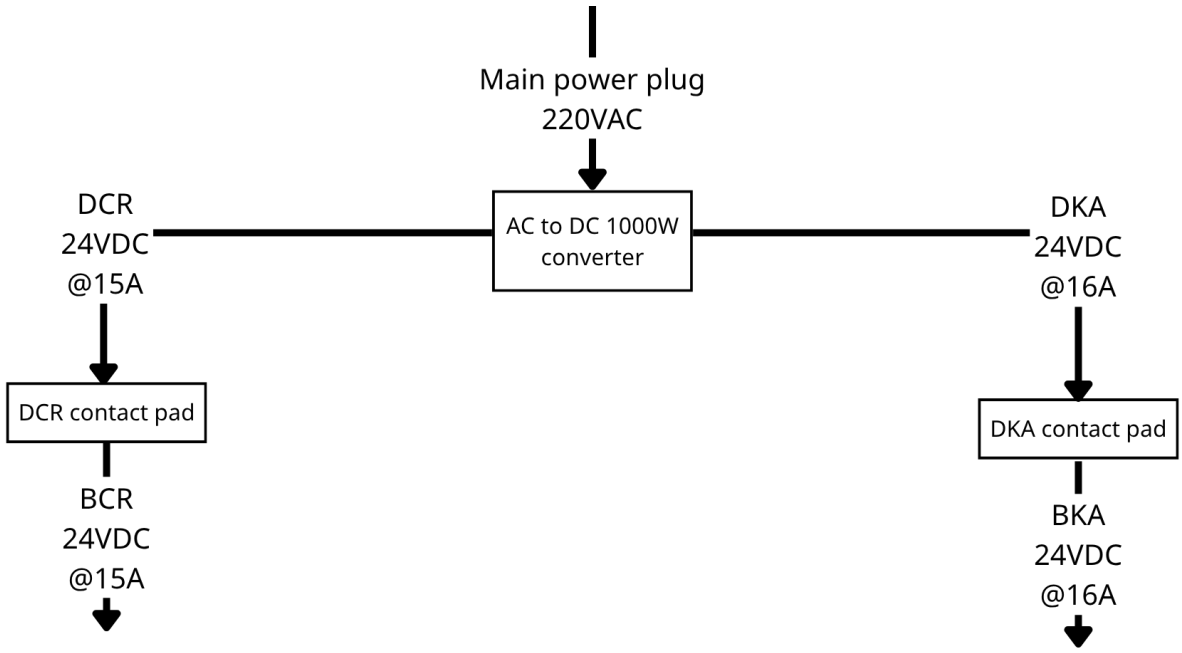
DKA	Dock side power rail to keep the entire system online while charging
DCR	Dock side charging rail
BBR	Bot side battery rail
BCR	Bot side charging rail
BKA	Bot side power rail connecting to DKA
BMR	Bot side main power rail

Source: The main power source consists of 3 LiFePO4 batteries rated at 15Ah, 3 in parallel, resulting in a total of 45Ah, which translates to roughly 1.5 hours of continuous motor activity.

Dock: The dock consists of 2 main 24VDC rails converted from 220VAC: DKA rail and DCR rail; DKA is at 24VDC@16A, whose main purpose is to offload the robot main rail (BMR) away from the battery rail (BBR**)**. The other rail, DCR, is set at 24VDC@15A and is used to charge the onboard batteries. The dock will use 3 copper rails that contact the corresponding 3 copper rails onboard, forming solid charging contacts. The layout is as follows: DKA on the outer right, GND in the middle, and DCR on the outer left, with 3cm spacing between them to help with electrical isolation. The charging protocol will follow a power ladder as follows: 5V@1A-12V@1A-24V@1A-24V@15A; designed to allow both DP and EM to cross-validate if the power rail order is correct and the voltage level is within a safety threshold of ±0.5V to ensure safe power recharging/handoff from the onboard. Disconnect BBR-BMR, connect DKA-BKA-BMR, cut BBR-BMR, and connect CR-BBR paths.



Dock's contact pad layout

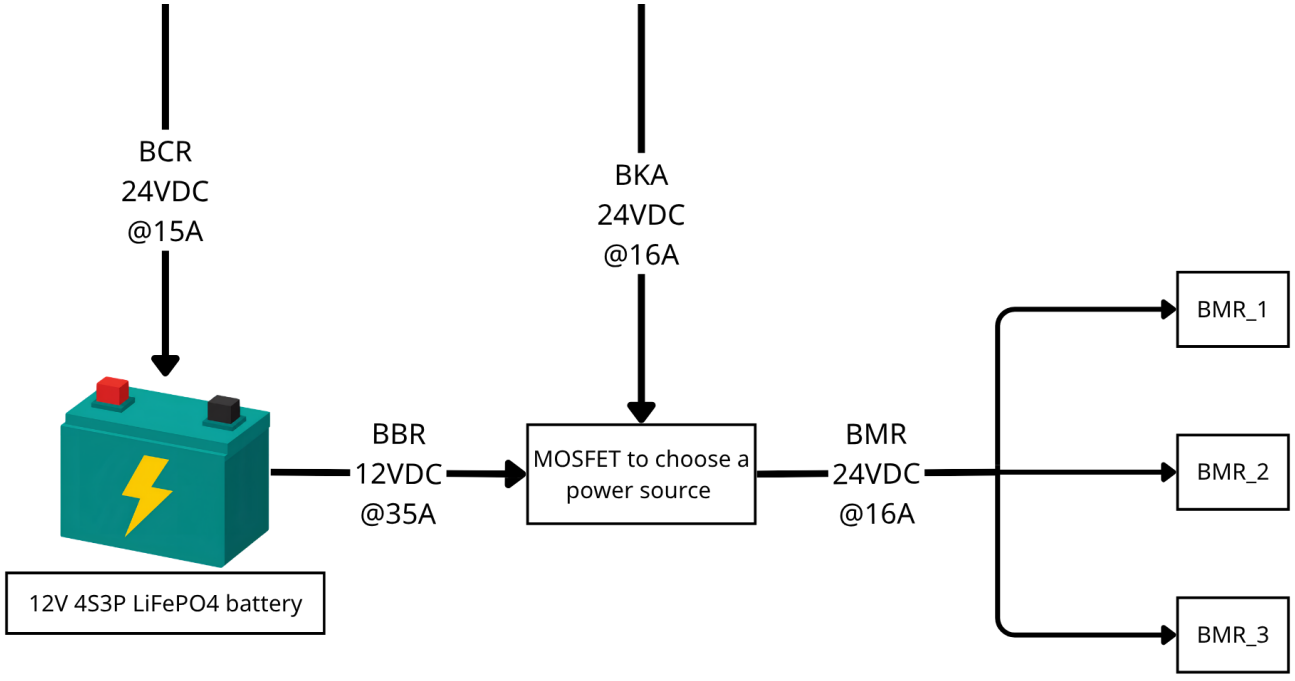


Basic layout of dock side power system

Robot:

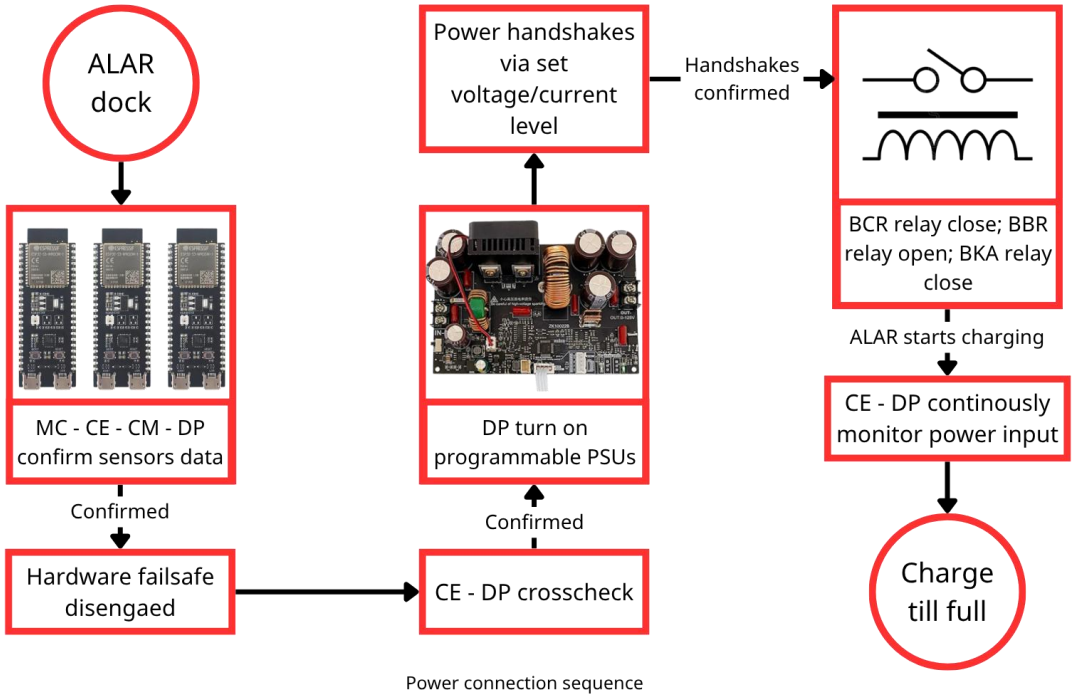
- **Battery:** The battery pack will provide 12VDC nominal at 32A, which will then be boosted to 24VDC@16A for BMR. BMR will then transmit the power to 3 different buck converters, which will reduce the voltage again to the corresponding rails. We have 3 different sub-power rails, denoted as BMR_1, BMR_2, and BMR_3. BMR_1 is responsible for the logic relays, fan power, and power sensors, as well as EM. BMR_2 is responsible for CM and close-range sensors, and lighting. BMR_3 is responsible for MC, LiDAR, cameras, and the display. There will be power sensors, fuses, and 1 18650 battery as an emergency source; this will ensure the safety of the power system against short circuits. To protect against voltage surges, high-voltage capacitors and isolation boards are present across important intersections, as well as power monitoring sensors and temperature sensors on each vital line connecting to EM. To protect against EM malfunctions, MC and CM can turn the relay on BBR - BMR off independently
- **Charging:** the battery pack is a combination of 3 separate LiFePO4 battery packs that all need 14.6V to fully charge. The charge rail CR is capable of 24VDC@15A, steps down to 14.6V to charge the battery, and the charge current will be 24A total, which calculates to be around 8A per pack, which leads to a theoretical full charge time of 2 hours from 0%. To ensure maximum safety as well as efficiency during the

charging phase, the use of a separate keep-alive rail KA is ideal; it will offload the battery to keep the vital systems online, able to charge at the highest speed possible, and, at the same time, protect the other components in case of a battery failure.



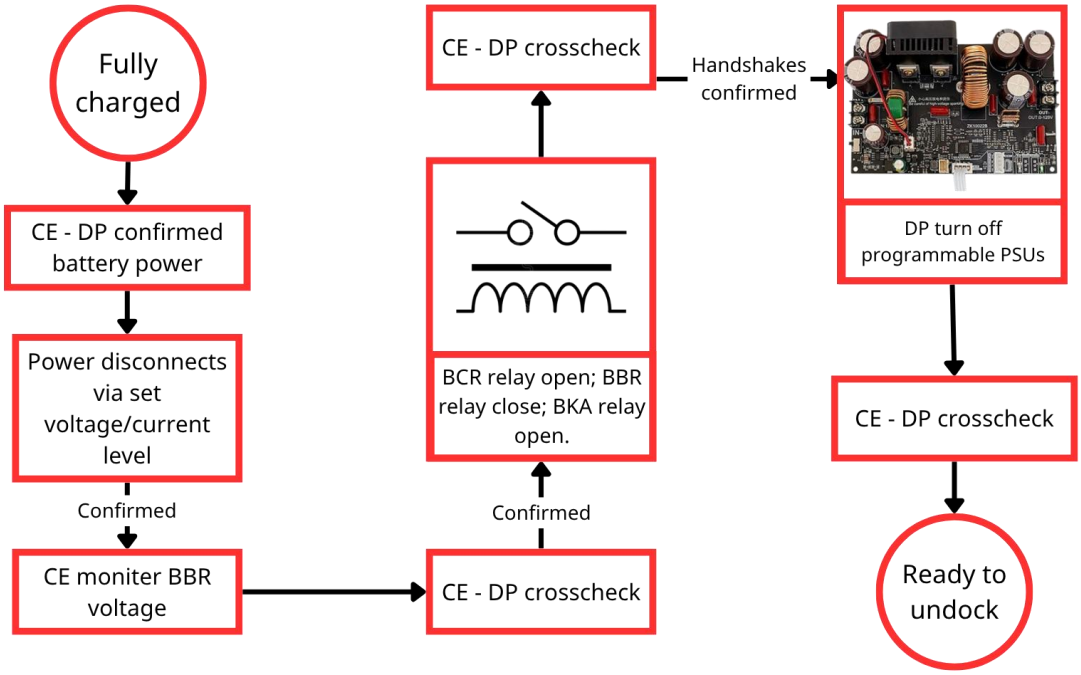
Basic layout of bot side power system

The power connection sequence is as follows:



Power connection sequence

The power disconnection sequence is as follows:



Power disconnection sequence

Navigation:

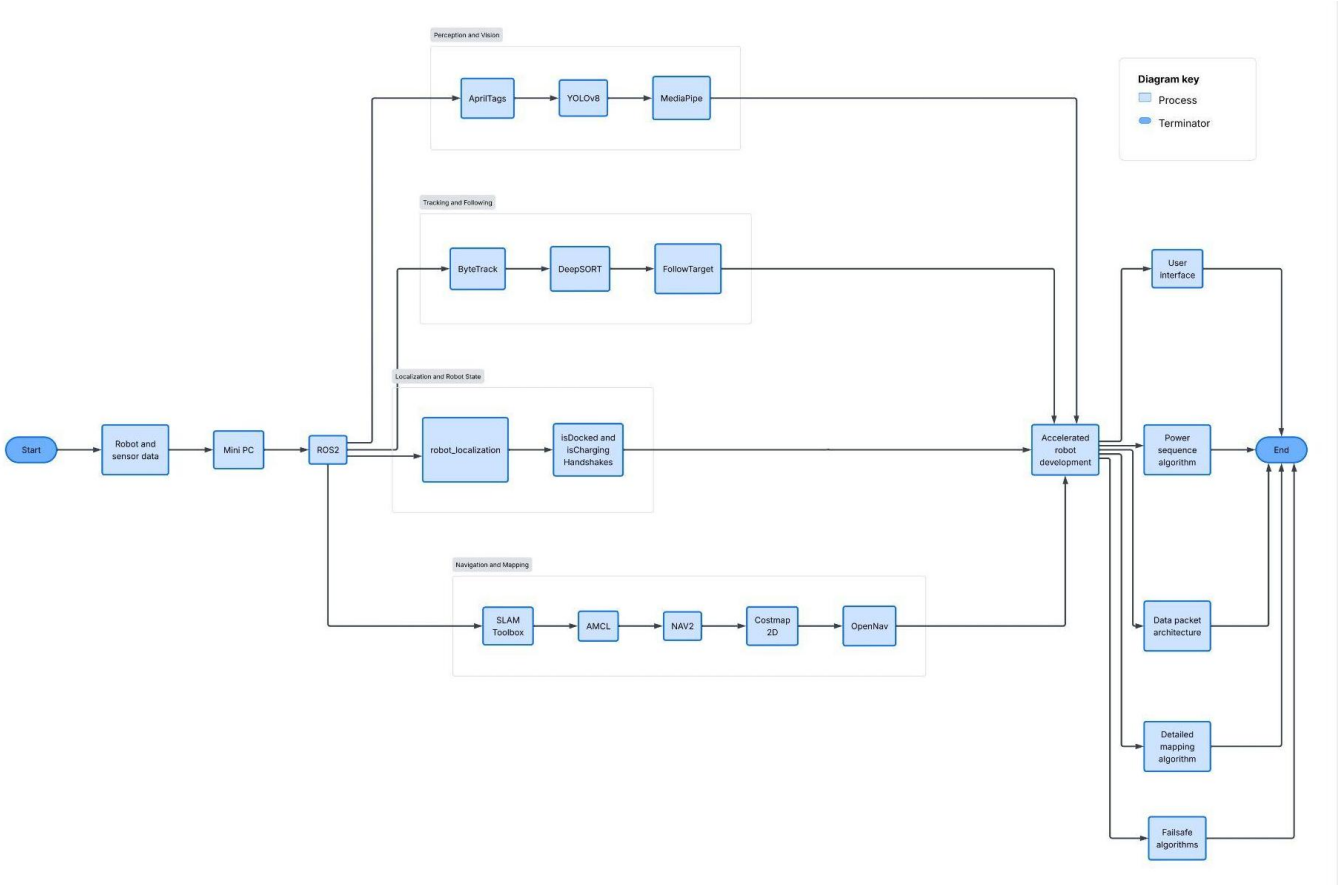
Hardware: ALAR makes use of a mini pc as the main navigation computer, combined with LiDAR for surrounding tracking; a front depth camera for object recognition and distance

analysis; a back camera for safety monitoring and docking; a webcam for people following and user interactions; IMUs for acceleration, tipping-over detection, and heading tracking; motor encoders to accurately track the path that has been made to recalibrate navigation systems.

Software: All the data will be fed into ROS2, a dedicated robot operating system, on the mini pc. It will then be split into different toolboxes like:

- SLAM toolbox
- AMCL
- NAV2
- AprilTags
- YOLOv8
- MediaPipe
- robot_localization
- OpenNav
- isDocked & isCharging Handshakes
- Costmap 2D
- ByteTrack
- DeepSORT
- FollowTarget

These toolboxes will make development easier, as well as save time to develop other things such as user interface, failsafe algorithms, data packet architecture, detailed mapping algorithm, and power sequence algorithm.



ROS2 Navigation Workflow

User Interaction (UI):

User Detection and Recognition: ALAR uses onboard cameras to detect users approaching the robot. When a user enters the detection area, the camera captures the user's image and provides visual information to the recognition system. This function allows ALAR to proactively assist ìf needed without requiring the user to operate the robot manually.

User Request and Communication: The robot can communicate with users through a visual display, and the user can provide a request to the robot, particularly when searching for a specific book or bookshelf. ALAR receives the user's request through its interaction interface and processes the provided information to determine the required service. The system then identifies the target book or location before starting the navigation process. This interaction is designed to reduce the number of actions required from the user when searching for books in the library.

Interface design: ALAR's interface follows principles of simplicity, friendliness, and usability. The system avoids unnecessary functions and focuses on the essential interactions needed to assist library users. Information presented on the display should be clear and easy

to understand.

Safety:

Mechanical: To give ALAR passive stability, we keep the battery and the heaviest components lower in the base to keep the center of gravity low to help prevent tipping over. Limiting the acceleration and turn rate in firmware as another defense line alongside hardware designs. Conduct load tests for the center multipurpose area to determine the safe operational limit weight. E-Stop button in an easy-to-reach location for hard shutdown in case of system malfunction.

Autonomy: If the communication between MC - CE - CM stops receiving a validated signal within a specific timeout, the system will autonomously stop the motors and disable the main power bus. Geofencing with QR-Tag to help the robot recalibrate its location and determine safe zones. Prevent bot recharge if MC - CE - DP - CM disagree on system status such as location on dock, sensor alignment, power rail validations, etc.

III. OUTCOME

The detail outcomes/products are as follows: (a prototype, a final report, attending a seminar....)

By the end of the research period, we will deliver a working prototype that fits into the library system. On its own, the robot will perform specific autonomous tasks, including:

- Guiding people to the desired area in the library.

- Acting as an autonomous shelf to assist with inventory management.

- Creating a dynamic map of the library layout daily, with the ability to create a heat map specializing in frequent/famous areas.

- The ability to self-sustain without extensive human intervention.

Apart from the prototype, a detailed technical report that documents the development process will be created. The report will describe the engineering parts of the robot such as the sensor-fusion navigation algorithm, the computer-vision pipeline, electrical and power-management systems, hardware architecture, and challenges we faced alongside with what we did to mitigate those challenges. This report will be open-sourced to give a reference and be a base for future robotics projects.

Finally, we designed the system with modularity in mind. By separating the core navigation and power systems from the library-specific vision component, the base architecture can be easily adapted for autonomous tasks across the university campus, such as document delivery or security patrols. This lets the robot technologies be reused or further developed in different applications.